Largest Range

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The problem

- [lo, hi] of the longest run of consecutive integers present in the array.

Intuition

- Only start counting at a number that has no predecessor, so each run is scanned exactly once.

Approach

- 1. Put all numbers in a set
- 2. For each number without number minus one, extend upward
- 3. Track the longest run's bounds

Java solution

```java
public class LargestRange {
    public static int[] largestRange(int[] a) {
        Set<Integer> s = new HashSet<>();
        for (int x : a) s.add(x);
        int[] best = {a[0], a[0]};
        for (int x : s) {
            if (s.contains(x - 1)) continue;
            int y = x;
            while (s.contains(y + 1)) y++;
            if (y - x > best[1] - best[0]) best = new int[]{x, y};
        }
        return best;
    }
}
```

Complexity

- Time: O(n)
- Space: O(n)

Recap

- Only start counting at a number that has no predecessor, so each run is scanned exactly once.
- Complexity: O(n)